**Interns KT to Complete Step-by-Step Jetty Installation on  EC2 (Port 9090)**

This guide helps students:

- Install Eclipse Jetty
- Deploy WAR application
- Change port to 9090
- Create welcome index page
- Access application using Public IP in Amazon Web Services EC2

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**Step 1 — Launch EC2 Instance**

Create Ubuntu EC2 instance.

Recommended:

- Ubuntu 22.04
- t2.micro

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**Step 2 — Configure Security Group**

Add inbound rules:

| Type | Port | Source |
|---|---|---|
| SSH | 22 | My IP |
| Custom TCP | 9090 | 0.0.0.0/0 |

This allows browser access to Jetty.

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**Step 3 — Connect to EC2**

ssh -i your-key.pem ubuntu@PUBLIC-IP

Example:

ssh -i test.pem ubuntu@44.204.200.50

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**Step 4 — Switch to Root User**

sudo su -

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**Step 5 — Update Server**

apt update -y
apt upgrade -y

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**Step 6 — Install Java**

Jetty requires Java.

apt install openjdk-17-jdk -y

Verify:

java -version

Expected:

openjdk version "17"

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**Step 7 — Create Jetty Directory**

mkdir /Jetty
cd /Jetty

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**Step 8 — Download Jetty**

wget https://repo1.maven.org/maven2/org/eclipse/jetty/jetty-home/11.0.20/jetty-home-11.0.20.tar.gz

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**Step 9 — Extract Jetty**

tar -xvf jetty-home-11.0.20.tar.gz

Verify:

ls

Expected:

jetty-home-11.0.20

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**Step 10 — Create Jetty Base Directory**

mkdir jetty-base

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**Step 11 — Initialize Jetty Base**

cd jetty-base
java -jar /Jetty/jetty-home-11.0.20/start.jar --add-modules=http,deploy

Expected folders:

resources
start.d
webapps

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**Step 12 — Change Port to 9090**

Open config:

vi start.d/http.ini

Find:

jetty.http.port=8080

Change to:

jetty.http.port=9090

Save and exit:

- ESC
- :wq

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**Step 13 — Create Welcome Root Application**

cd webapps
mkdir root
cd root

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**Step 14 — Create Index File**

vi index.html

Add:

```
<html>
<head>
<title>Jetty Application</title>
</head>

<body>
<h1>Welcome to Jetty Server</h1>
<h2>Running on AWS EC2</h2>
<p>DevOps Training Demo</p>
</body>
</html>
```

Save:

- ESC
- :wq

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**Step 15 — Deploy WAR File**

Copy WAR file into webapps folder.

Example:

cd ..

Now folder should look like:

ls

Expected:

root  studentapp.war

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**Step 16 — Start Jetty Server**

IMPORTANT:
Start from jetty-base directory only.

cd /Jetty/jetty-base
java -jar /Jetty/jetty-home-11.0.20/start.jar

Expected:

Started Server

DO NOT close terminal.

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**Step 17 — Verify Port Listening**

Open another terminal.

Check:

ss -tulnp | grep 9090

Expected:

LISTEN 0 50 *:9090

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**Step 18 — Test Inside Server**

wget http://localhost:9090

Should download index page successfully.

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**Step 19 — Access From Browser**

Open:

http://PUBLIC-IP:9090

Example:

http://44.204.200.50:9090

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**Step 20 — Access WAR Application**

http://PUBLIC-IP:9090/studentapp

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**Run Jetty in Background (Optional)**

Instead of foreground:

nohup java -jar /Jetty/jetty-home-11.0.20/start.jar &

Check process:

ps -ef | grep java

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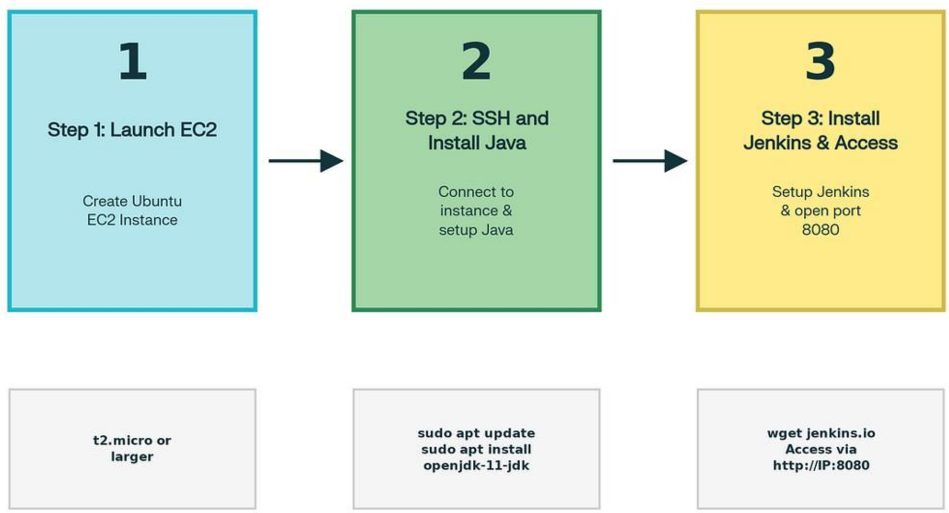
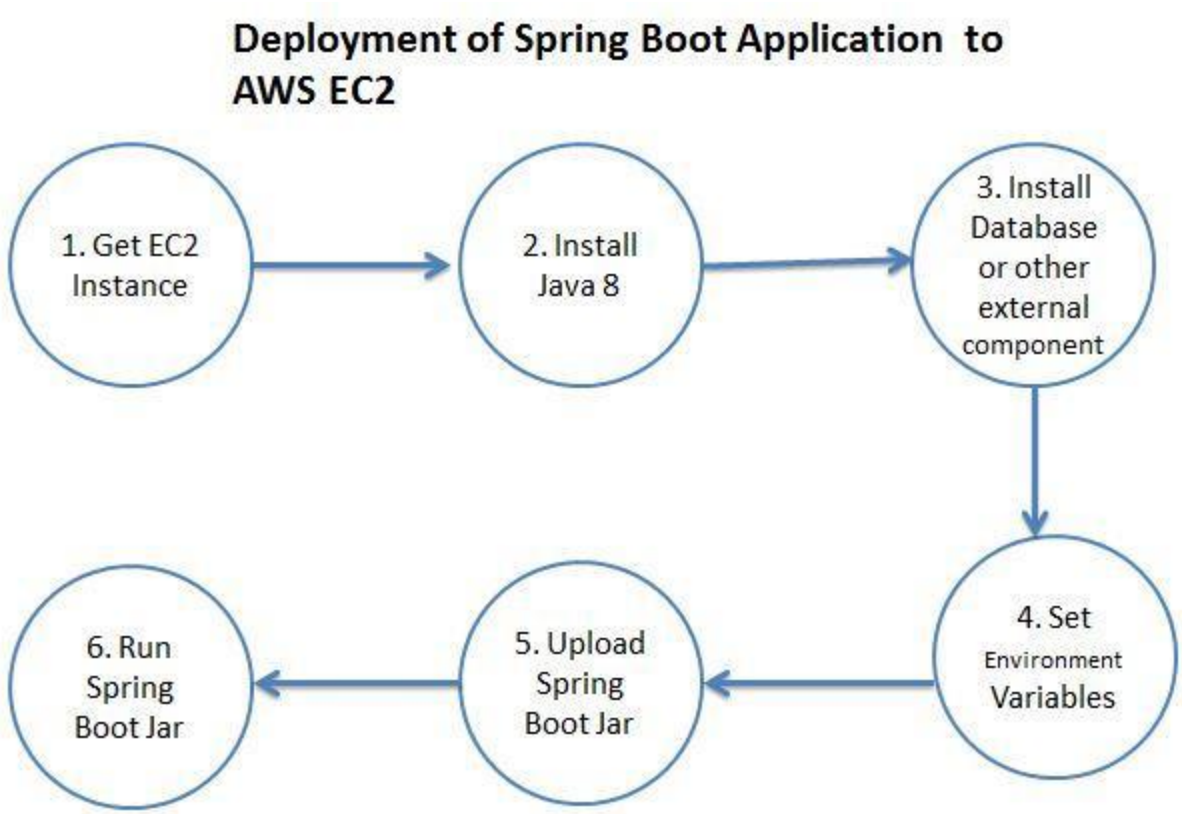
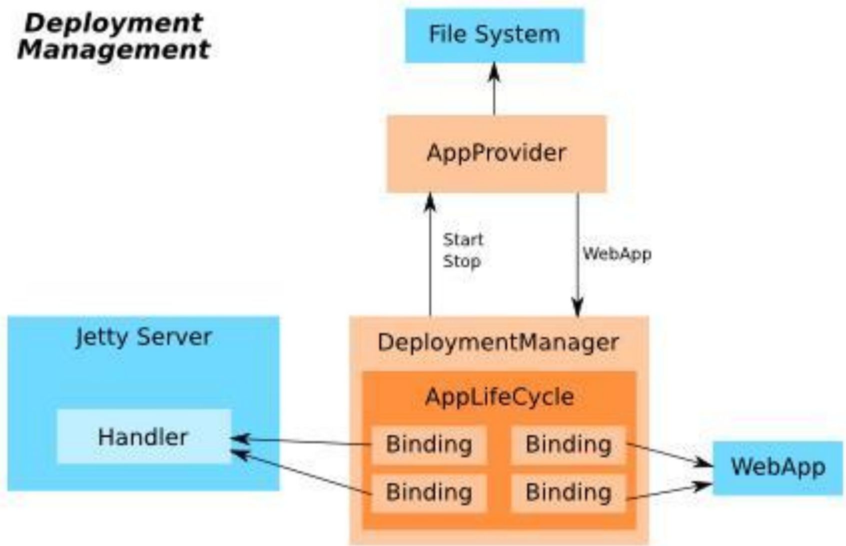
**Architecture Diagram**


1
Step 1: Launch EC2
Create Ubuntu
EC2 Instance
2
Step 2: SSH and
Install Java
Connect to
instance &
setup Java
3
Step 3: Install
Jenkins & Access
Setup Jenkins
& open port
8080
t2.micro or
larger
sudo apt update
sudo apt install
openjdk-11-jdk
wget jenkins.io
Access via
http://IP:8080


Deployment of Spring Boot Application to
AWS EC2
1. Get EC2
Instance
2. Install
Java 8
3. Install
Database
or other
external
component
4. Set
Environment
Variables
5. Upload
Spring
Boot Jar
6. Run
Spring
Boot Jar

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**Important Concepts for Students**

**Jetty Home**

Contains:

- binaries
- libraries
- start.jar

Example:

/Jetty/jetty-home-11.0.20

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**Jetty Base**

Contains:

- applications

- configs
- deployments

Example:

/Jetty/jetty-base